Introduction

The ER (Entity-Relationship) diagram represents the structure of a **Cab Booking System**, similar to platforms like Uber or Ola. It models how different components of the system—such as riders, drivers, cabs, trips, and payments—interact with each other. The diagram helps in designing the database by clearly defining entities, their attributes, and the relationships between them.

Entities and Their Attributes

1. RIDER

- rider_id (PK), full_name, phone, rating, created_at
- Represents users who book rides.

2. RIDE_REQUEST

- request_id (PK), requested_at, pickup, drop_point, distance_km, status
- Stores ride booking requests.

3. TRIP

- trip_id (PK), start_time, end_time, state, fare
- Represents an actual ride after request acceptance.

4. DRIVER

- driver_id (PK), full_name, phone, license_no, status, rating, created_at
- Represents drivers providing the service.

5. CAB

- cab_id (PK), car_number, car_type, seats, is_active
- Represents vehicles used for rides.

6. PAYMENT

- payment_id (PK), amount, method, payment_state, paid_at
- Stores payment details for trips.

Relationships and Their Types

1. RIDER → RIDE_REQUEST

- Relationship**: REQUESTS**
- **Type: 1 : N (One-to-Many)**
- One rider can create multiple ride requests, but each request belongs to one rider.

2. RIDE_REQUEST → TRIP

- Relationship**: GENERATES**
- **Type: 1 : 1 (One-to-One)**
- Each ride request generates exactly one trip.

3. TRIP → CAB

- Relationship**: USES**
- **Type: N : 1 (Many-to-One)**
- Many trips can use the same cab, but each trip uses only one cab.

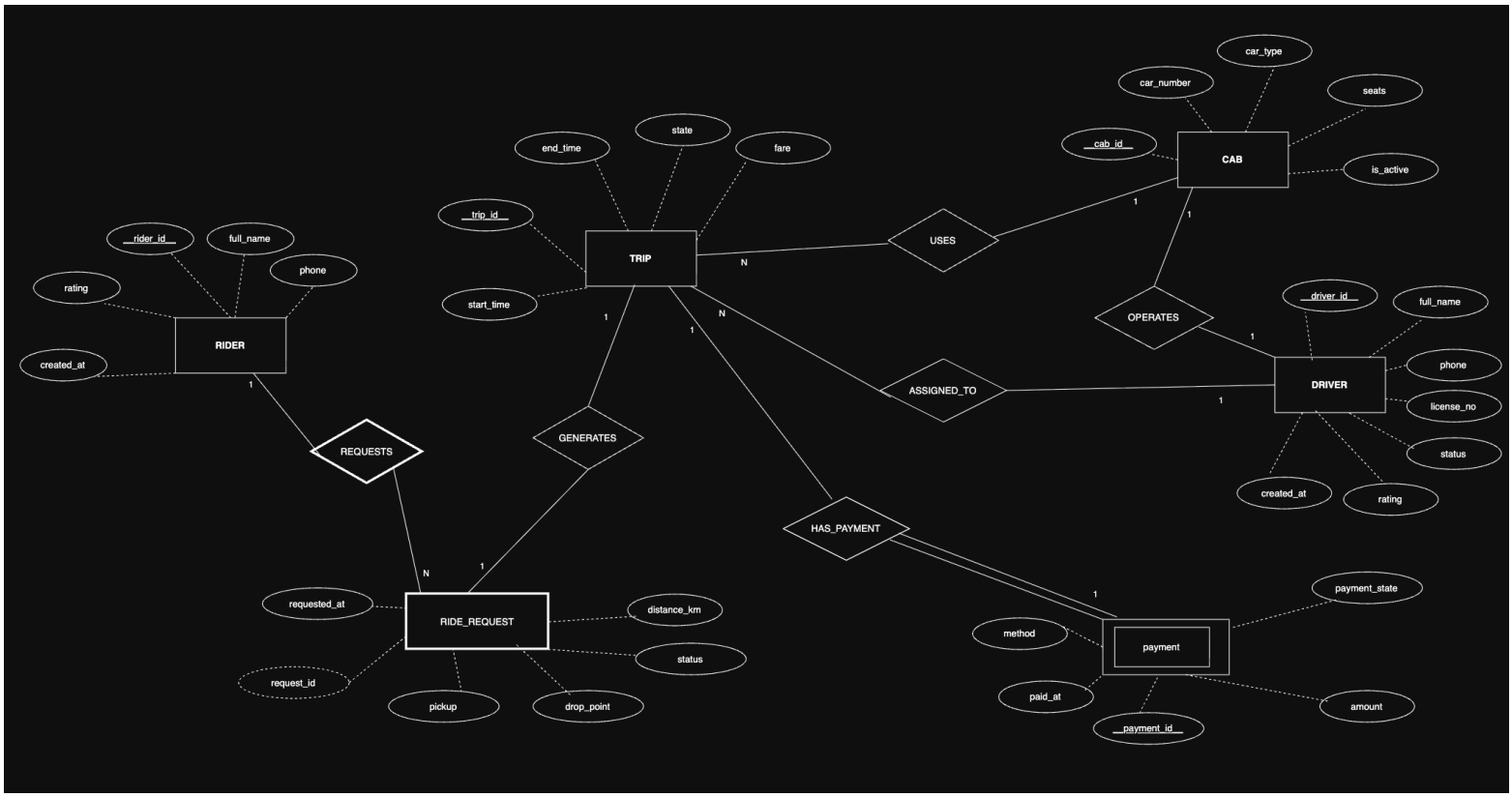
4. TRIP → DRIVER

- Relationship**: ASSIGNED_TO**
- **Type: N : 1 (Many-to-One)**
- Each trip is assigned to one driver, but a driver can handle multiple trips.

5. DRIVER → CAB

- Relationship**: OPERATES**
- **Type: 1 : 1 (One-to-One)** *(as per diagram)*
- Each driver operates one cab, and each cab is operated by one driver.

6. TRIP → PAYMENT

- Relationship**: HAS_PAYMENT**
- **Type: 1 : 1 (One-to-One)**
- Each trip has exactly one associated payment.

Entity Relationship Diagram :-